



Topics in this chapter:

1. Temperature Scales & Thermometers
2. Nuclear Physics & Atomic Discoveries
3. Electricity, Magnetism & Electromagnetism
4. Optical Instruments & Waves
5. Gravitation, Motion & Mechanics
6. Nobel Prizes & Awards
7. Miscellaneous Inventions
8. ⚠ Traps / Key Differences Summary (Q1471–Q1489)

1. Temperature Scales & Thermometers

- Daniel Gabriel Fahrenheit ne modern mercury-in-glass thermometer aur Fahrenheit scale ka invention kiya; alcohol thermometer bhi inhone banaya. [Q1471]
- Anders Celsius ne Celsius scale invent kiya; Royal Society ke Fellow the. [Q1471]
- Galileo Galilei ne telescope ki khoj ki thi; thermometer se directly jude nahi hain. [Q1471]

2. Nuclear Physics & Atomic Discoveries

- Albert Einstein ne Theory of Relativity di; formula $E = mc^2$ (mass-energy equivalence). [Q1472]
- John Dalton ne Atomic Theory di. [Q1472]
- Christian Bernard South African cardiac surgeon the — pehla human-to-human heart transplant kiya. [Q1472]
- Thomas Alva Edison ne light bulb invent kiya. [Q1472]
- Ernest Rutherford ne proton discover kiya; nuclear model of atom diya. [Q1475]
- J.J. Thomson ne electron discover kiya (cathode ray experiment). [Q1475]
- James Chadwick ne neutron discover kiya (1932). [Q1475]
- Niels Bohr ne Bohr model diya — electrons definite orbits mein ghumte hain. [Q1475]
- Otto Hahn ne nuclear fission discover kiya. [Q1475]
- Marie Curie ne polonium aur radium discover kiye; Nobel Prize jeeta. [Q1478]
- Enrico Fermi ne nuclear reactor banaya; controlled chain reaction. [Q1480]
- Henri Becquerel ne radioactivity discover ki. [Q1481]
- Werner Heisenberg ne Uncertainty Principle diya. [Q1483]
- Max Planck ne Quantum Theory di; energy packets (quanta) ka concept. [Q1484]
- A.P.J. Abdul Kalam ko "Missile Man of India" kehte hain. [Q1487]
- C.V. Raman ne Raman Effect discover kiya; Nobel Prize in Physics (1930) jeeta. [Q1489]

3. Electricity, Magnetism & Electromagnetism

- Hans Christian Oersted ne electromagnetism discover kiya; electric current se magnetic field hoti hai. [Q1473]

- Michael Faraday ne electromagnetic induction discover kiya; dynamo/generator principle. [Q1474]
- James Clerk Maxwell ne electromagnetic waves ki theory di. [Q1474]
- André-Marie Ampère ne current-carrying conductor ke magnetic field ka study kiya. [Q1473]
- Nikola Tesla ne AC motor/generator ka kaam kiya. [Q1474]
- Alessandro Volta ne electric battery invent ki. [Q1476]
- Georg Simon Ohm ne Ohm's Law di; resistance ka unit "Ohm" inke naam par. [Q1477]
- Benjamin Franklin ne lightning rod invent kiya; electricity ka nature samjha. [Q1479]

4. Optical Instruments & Waves

- Hans Lippershey ne telescope invent kiya (1608). [Q1482]
- Galileo Galilei ne telescope ko astronomy mein use kiya; Jupiter ke moons dekhe. [Q1482]
- Antonie van Leeuwenhoek ne microscope ko improve kiya; bacteria dekhe. [Q1482]
- Guglielmo Marconi ne radio invent kiya; wireless telegraphy. [Q1485]
- Alexander Graham Bell ne telephone invent kiya (1876). [Q1485]
- Heinrich Hertz ne electromagnetic waves experimentally demonstrate kiya. [Q1485]

5. Gravitation, Motion & Mechanics

- Sir Isaac Newton ne teen major contributions diye: Laws of Motion (3 laws), Law of Universal Gravitation, aur Calculus (Leibniz ke saath). [Q1474, Q1486]
- Albert Einstein ne General Relativity di — gravitation ko spacetime curvature se explain kiya. [Q1486]
- Galileo Galilei ne motion aur telescope pe kaam kiya. [Q1486]

6. Nobel Prizes & Awards

- C.V. Raman — Nobel Prize in Physics (1930) — Raman Effect. [Q1489]
- Marie Curie — 2 Nobel Prizes (Physics 1903, Chemistry 1911). [Q1478]
- Albert Einstein — Nobel Prize in Physics (1921) — Photoelectric Effect. [Q1472]
- Max Planck — Nobel Prize in Physics (1918). [Q1484]
- Enrico Fermi — Nobel Prize in Physics (1938). [Q1480]
- Werner Heisenberg — Nobel Prize in Physics (1932). [Q1483]

7. Miscellaneous Inventions

- Wright Brothers (Orville & Wilbur) ne aeroplane invent kiya (1903). [Q1488]
- A.P.J. Abdul Kalam — Missile Man of India; DRDO/ISRO mein kaam. [Q1487]
- Benjamin Franklin — lightning rod, bifocal glasses, Franklin stove. [Q1479]
- Guglielmo Marconi — radio; wireless communication ka father. [Q1485]

8. ⚠ Traps / Key Differences Summary (Q1471–Q1489)

- ⚠ Trap: Fahrenheit ne mercury thermometer banaya; Celsius ne Celsius scale — donon alag hain. [Q1471]
- ⚠ Trap: Einstein relativity ke liye famous hain lekin Nobel Prize Photoelectric Effect ke liye mila tha. [Q1472]
- ⚠ Trap: Rutherford ne proton diya; Thomson ne electron; Chadwick ne neutron — teenon alag hain. [Q1475]
- ⚠ Trap: Faraday induction ke liye; Maxwell waves ki theory ke liye; Oersted current-magnetism link ke liye. [Q1473, Q1474]
- ⚠ Trap: Galileo ne telescope use kiya astronomy mein lekin invent Lippershey ne kiya tha. [Q1482]
- ⚠ Trap: Newton classical gravitation; Einstein general relativity — donon alag theories hain. [Q1486]
- ⚠ Trap: C.V. Raman Indian scientist hain jo Nobel jeete; ye fact exam mein direct pucha jaata hai. [Q1489]